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Double Debiased Machine Learning Nonparametric Inference with Continuous Treatments

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ABSTRACT

We propose a doubly robust inference method for causal effects of continuous treatment variables, under unconfoundedness and with nonparametric or high-dimensional nuisance functions. Our double debiased machine learning (DML) estimators for the average dose–response function (or the average structural function) and the partial effects are asymptotically normal with nonparametric convergence rates. The first-step estimators for the nuisance conditional expectation function and the conditional density can be nonparametric or ML methods. Using a kernel-based doubly robust moment function and cross-fitting, we give high-level conditions under which the nuisance function estimators do not affect the first-order large sample distribution of the DML estimators. We provide sufficient low-level conditions for kernel, series, and deep neural networks. We justify the use of kernel to localize the continuous treatment at a given value by the Gateaux derivative. We implement various ML methods in Monte Carlo simulations and an empirical application on a job training program evaluation.

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Average structural function;
Cross-fitting; Dose–response
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1. Introduction

We propose a nonparametric inference method for *continuous* treatment or structural effects. We focus on the heterogeneous effect with respect to the continuous treatment or policy variable T . To identify the causal effects, it is plausible to have a large number of control variables X such that T is randomly assigned conditional on X . To employ machine learning (ML) methods for nuisance functions and to achieve doubly robust inference, we use a double debiased ML approach that combines a doubly robust moment function and cross-fitting.

We consider a nonparametric and model-free outcome equation $Y = g(T, X, \varepsilon)$. No functional form assumption is imposed on the unobserved disturbances ε , such as restrictions on dimensionality, monotonicity, or separability. The potential outcome is $Y(t) = g(t, X, \varepsilon)$ indexed by the hypothetical treatment value t . The object of interest is the *average dose–response function* as a function of t , defined as the expected value of the potential outcome across observations with the observed and unobserved heterogeneity (X, ε) , that is, $\beta_t \equiv \mathbb{E}[Y(t)] = \int \int g(t, X, \varepsilon) dF_{X\varepsilon}$. It is also known as the *average structural function* in nonseparable models in Blundell and Powell (2003). The well-studied average treatment effect of switching from treatment t to s is $\beta_s - \beta_t$. We allow the continuous treatment to be multi-dimensional and hence capture the unrestricted heterogeneous effects with respect to the multivariate treatment

variable. We define the *partial (or marginal) effect* of the first component of the continuous variable T at $t = (t_1, \dots, t_{d_T})'$ to be the partial derivative $\theta_t \equiv \partial \beta_t / \partial t_1$.¹ In program evaluation, the average dose response function β_t shows how participants' labor market outcomes vary with the length of exposure to a job training program. In demand analysis when T contains price and income, the average structural function β_t can be the Engel curve. The partial effect θ_t reveals the average price elasticity at given values of price and income. Other examples include the efficacy of political advertisements on campaign contributions in Fong, Hazlett, and Imai (2018), the effect of nurse staffing on hospital readmissions penalties in Kennedy et al. (2017), etc.

The average structural function β_t and the partial effect θ_t of continuous variables are *non-regular nonparametric* objects which cannot be estimated at a root- n convergence rate. We propose a kernel-based double debiased ML (DML) estimator that uses two key features: (i) a doubly robust/orthogonal moment function to reduce sensitivity in estimating β_t with respect to nuisance functions, and (ii) *cross-fitting* via sample-splitting to remove bias induced by overfitting and achieve stochastic equicontinuity without strong entropy conditions: (i) The moment function is

$$\gamma(t, X_i) + \frac{K_h(T_i - t)}{f_{T|X}(t|X_i)} (Y_i - \gamma(t, X_i)), \quad (1.1)$$

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¹In contrast, the partially linear model $Y = \theta T + g(X) + \varepsilon$ in Robinson (1988) specifies the homogenous effect by θ and hence is a semiparametric problem. The important partially linear model has many applications and is one of the leading examples in the recent ML literature; see Chernozhukov et al. (2018) and references therein. Another semiparametric parameter of interest is the weighted average of β_t or θ_t over a range of treatment values t , such as the average derivative that summarizes certain aggregate effects (Powell, Stock, and Stoker 1989).

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where the conditional expectation function $\gamma(t, x) \equiv \mathbb{E}[Y|T = t, X = x]$ and the conditional density $f_{T|X}(t|x)$ is also known as the generalized propensity score (GPS). A kernel $K_h(T_i - t)$ weights observation i with treatment value around t in a distance of h . The number of such observations shrinks as the bandwidth h vanishes with the sample size n .

(ii) A L -fold cross-fitting splits the sample into L subsamples. The nuisance function estimators for $\gamma(t, X_i)$ and $f_{T|X}(t|X_i)$ use observations in the other $(L - 1)$ subsamples that do not contain the observation i . Then the DML estimator $\hat{\beta}_t$ for β_t is the sample average of the estimated moment function (1.1) over the full sample. We estimate the partial effect θ_t by a numerical differentiation of $\hat{\beta}_t$.

We provide high-level rate conditions on the nuisance function estimators for $\gamma(t, X_i)$ and $f_{T|X}(t|X_i)$ such that the proposed DML estimators are asymptotically normal, and the nuisance function estimators do not affect the first-order asymptotic distribution. Therefore, we can estimate the nuisance functions by a variety of nonparametric and ML methods which satisfy such high-level conditions. As each ML method has its strengths and weaknesses depending on the data generating process and applications, flexible employment of various nuisance function estimators is desirable.

Each nuisance function which satisfies the high-level rate conditions has its specific low-level conditions on smoothness, tuning parameters, dimensionality, etc. For low(fixed)-dimensional control variables X , we give the low-level conditions for nonparametric kernel estimators, series estimators, and the deep neural networks in Farrell, Liang, and Misra (2021b) that is widely popular in industrial applications and useful for semiparametric inference on the average binary treatment effect as illustrated in Farrell, Liang, and Misra (2021b). Our high-level conditions on the convergence rates use a *partial L_2 norm* that fixes the treatment value at t , in contrast to the standard L_2 norm that integrates over the joint distribution of (T, X) , that is, the root-mean-square rate. Our results on the convergence rates of these nuisance function estimators are new. In high-dimensional settings where the dimension of X grows with the sample size, our inference theory is valid as long as the nuisance function estimators satisfy the high-level rate conditions, for example, Lasso.

Importantly our inference theory for the proposed DML estimator allows one of the nuisance functions to be misspecified. Specifically the standard errors using our asymptotic theory are robust to such misspecification to construct valid confidence intervals for β_t , which is a model-based causal parameter. Such doubly robust inference is useful, especially when the nuisance functions are estimated under a parametric model or sparsity approximation, such as Lasso.

Furthermore, we propose a novel Simulated DML (SDML) estimator that replaces $\gamma(t, X_i)$ in (1.1) with $\gamma(U_i, X_i)$ where a simulated variable U_i localizes the realized treatment values around the target value t . Introducing such a local variation allows us to characterize the high-level rate conditions on the nuisance function estimator using the standard L_2 norm that is available for nonparametric kernel and series estimators, as well as recent ML estimators, such as Lasso (Bickel, Ritov, and Tsybakov 2009), neural networks (Chen and White 1999;

Schmidt-Hieber 2020; Farrell, Liang, and Misra 2021b), random forests (Syrkanis and Zampetakis 2020), and empirical L_2 rate for boosting (Luo and Spindler 2016), as discussed in Chernozhukov et al. (2018) and Chernozhukov, Newey, and Singh (2022a).² This result is valuable in practice, as the SDML estimator readily allows for a wider class of ML methods. In contrast, the DML estimator assumes the high-level rate conditions using a partial L_2 norm that is not directly available in the literature. We also provide the standard L_2 -rate condition for the DML estimator at the cost of losing the doubly robust inference and stricter regularity conditions.

In addition, we propose generic ML estimators for the *reciprocal of the conditional density* $1/f_{T|X}(t|x)$ when with one continuous treatment ($d_T = 1$) and for $f_{T|X}(t|x)$ when $d_T \geq 1$, respectively, which may be of independent interest. We also propose a data-driven bandwidth to consistently estimate the optimal bandwidth that minimizes the asymptotic mean squared error.

Related literature: We are among the first to apply the DML approach to inference on the average structural function β_t and the partial effect θ_t of continuous variables, to our knowledge. The doubly robust moment function in (1.1) has appeared in Kallus and Zhou (2018) and Su, Ura, and Zhang (2019).³ Kallus and Zhou (2018) consider a counterfactual treatment by a known policy $\tau(X_i)$ and their object of interest is $\mathbb{E}[Y_i(\tau(X_i))]$. Their asymptotic theory assumes a known GPS that is not estimated. Su, Ura, and Zhang (2019) focus on Lasso-type estimators. In contrast, we use cross-fitting and provide high-level rate conditions on the two nuisance function estimators, which facilitate a variety of nonparametric and ML methods. Moreover, our estimator is doubly robust on inference, that is, our inference theory and the asymptotic distribution accommodate the case when either one of the nuisance functions $\mathbb{E}[Y|T, X]$ or $f_{T|X}$ is misspecified, as in Kennedy et al. (2017) and Takatsu and Westling (2024). This is a stronger result than the usual doubly robustness on the consistency of the estimator.

We further discuss in more details the benefit of cross-fitting, which has been widely used in the recent ML literature and advocated in Chernozhukov et al. (2018) (CCDDHNR, hereafter). Cross-fitting eliminates the need for Donsker conditions for the nuisance function estimators, as in conventional semiparametric statistical and econometric analysis. This is key for flexibly estimating the nuisance functions, because Donsker conditions are unknown for many ML methods, while their convergence rates are better understood. More importantly, such commonly used results in empirical process theory are known to be inappropriate in some high-dimensional settings, where the nuisance function estimators necessarily live in highly complex spaces as the dimension of X increases. We refer to CCDDHNR for detailed discussion on the sparse high-dimensional IV model

²We are grateful to Whitney Newey for the idea of simulating U_i from the probability density function $f_U(u) = f_T((u - t)/h)/h$.

³The first version of this article is circulated in 2019. The rapidly growing ML literature has used the double robustness to reduce regularization and modeling biases in estimating the nuisance functions by ML or nonparametric methods; for example, Belloni, Chernozhukov, and Hansen (2014), Farrell (2015), Belloni et al. (2017), Farrell, Liang, and Misra (2021b), Chernozhukov et al. (2022), Chernozhukov et al. (2018), Rothe and Firpo (2019), and references therein.

in Belloni et al. (2012) and the unconditional treatment effects in Belloni et al. (2017).⁴ Next we discuss Fan et al. (2022) as a specific example to illustrate how cross-fitting weakens the sparsity assumption in the Lasso methods.

Fan et al. (2022) estimate the conditional average binary treatment effect (CATE) $\mathbb{E}[Y(1) - Y(0)|X_1]$ for a low-dimensional subset X_1 of X , using an orthogonal moment function. Fan et al. (2022) study the cross-fitting estimator and the full-sample estimator without sample-splitting. When the nuisance functions are estimated by Lasso methods, they derive explicitly that the cross-fitting estimator requires substantially weaker sparsity conditions than the full-sample estimator. They also find that the cross-fitting estimator has better finite-sample properties. As their cross-fitting estimator takes a similar form as our DML estimator, their findings on the benefit of cross-fitting should carry over here. Furthermore, our doubly robust inference alleviates the concern of misspecification and sparsity conditions, which has not been explored in the prior literature on Lasso.

Our work builds on the results for semiparametric models in Ichimura and Newey (2022), Chernozhukov et al. (2022), and CCDDHNR and extends the literature to nonparametric continuous treatment/structural effects. Note that the doubly robust estimator for a binary/multivalued treatment replaces the kernel $K_h(T_i - t)$ with the indicator function $\mathbf{1}\{T_i = t\}$ in equation (1.1) and has been widely studied. The advantageous properties of the DML estimator for the binary treatment carry over to the continuous treatments case. While our kernel-based estimator appears to be a simple modification of the binary treatment case, we show that one important distinct feature of *non-regular nonparametric* parameters is that the Gateaux derivative and the Riesz representer are not unique, depending on how we approximate the continuous treatment distribution approaching a point mass. And the kernel function is a natural choice for localization at t . Neyman (1959) orthogonality holds as $h \rightarrow 0$. Therefore, we provide a foundational justification for the proposed kernel-based DML estimator, relative to alternative approaches, such as Kennedy et al. (2017), Semenova and Chernozhukov (2020), van der Laan, Bibaut, and Luedtke (2018).

There is a small yet rapidly growing literature on employing the DML approach for non-regular nonparametric infinite-dimensional objects. Hsu et al. (2024) propose a test for monotonicity of β_t . Chernozhukov, Newey, and Singh (2022b), Semenova and Chernozhukov (2020), Fan et al. (2022), and Zimmert and Lechner (2019) study the CATE of a binary treatment given some low-dimensional X_1 . Their objects of interest are different from our average structural function and partial effect of continuous treatments. As our DML estimators, most of the papers mentioned above use a kernel to localize X_1 , while Semenova and Chernozhukov (2020) use a series-based localization. Despite the advantageous theoretical properties, we note potential drawbacks of the DML approach: Double robustness often requires additional nuisance function estimation, such as the conditional density in the Riesz representer, which could introduce additional variation and

implementation complication. Cross-fitting could result in a small effective sample size.

Our article adds to the literature on continuous treatment effects. In low-dimensional settings, see Imbens (2000), Hirano and Imbens (2004), Flores (2007), and Lee (2018) for examples of a class of regression-type estimators $n^{-1} \sum_{i=1}^n \hat{\gamma}(t, X_i) = n^{-1} \sum_{i=1}^n \hat{\mathbb{E}}[Y|T = t, X = X_i]$ and $n^{-1} \sum_{i=1}^n \hat{\mathbb{E}}[Y|T = t, \hat{f}_{T|X}(t|X) = \hat{f}_{T|X}(t|X_i)]$. Galvao and Wang (2015) and Hsu et al. (2020) study a class of inverse probability weighting estimators. The empirical applications in Flores et al. (2012) and Kluve et al. (2012) focus on semiparametric results. We extend the literature to a DML framework that enables ML methods for nonparametric inference in practice. In high-dimensional settings with penalization approaches like Lasso, we may use the additive model in Meier, van de Geer, and Bühlmann (2009), assuming $Y = f_1(T) + f_2(X) + \varepsilon$, then $\beta_t = f_1(t)$ under a normalization $\mathbb{E}[f_2(X) + \varepsilon] = 0$.

The article proceeds as follows. Section 2 introduces the framework and estimation procedure. Section 3 presents the asymptotic theory of the DML estimators and low-level conditions for various nuisance function estimators. Section 4 introduces the Simulated DML estimator. Section 5 demonstrates the usefulness of our DML estimator with various ML methods in Monte Carlo simulations and an empirical example on the Job Corps program evaluation. The supplementary appendix presents all the proofs and additional results, such as uniform inference theory.

2. Setup and Estimation

Let $\{Y_i, T_i, X_i\}_{i=1}^n$ be an iid sample from $Z = (Y, T', X')' \in \mathcal{Z} = \mathcal{Y} \times \mathcal{T}_0 \times \mathcal{X} \subseteq \mathcal{R}^{1+d_T+d_X}$ from a population P with a cumulative distribution function (CDF) $F_Z(Z)$. Consider a set of treatment values of interest $\mathcal{T}$ to be an interior of the support of T .

Assumption 2.1. (a) (Conditional independence) T and ε are independent conditional on X ; (b) (Common support) $\inf_{t \in \mathcal{T}} \text{ess inf}_{x \in \mathcal{X}} f_{T|X}(t|x) \geq c$ for some positive constant c^5 ; (c) $f_Z(y, t, x)$ and $\mathbb{E}[Y|T = t, X = x]$ are three-times differentiable with respect to (w.r.t.) t with all three derivatives being bounded uniformly over $(y, t', x')' \in \mathcal{Z}$; (d) $\text{var}(Y|T = t, X = x)$ and its derivatives w.r.t. t are bounded uniformly over $\mathcal{T} \times \mathcal{X}$.

The commonly used identifying Assumption 2.1(a) based on observational data (also known as unconfoundedness, selection on observables, or ignorability) assumes that conditional on observables, the treatment variable is as good as randomly assigned, or conditionally exogenous.⁶

⁵The essential infimum is defined as the supremum of the essential lower bounds, $\text{ess inf}_{x \in \mathcal{X}} f_{T|X}(t|x) = \sup\{C \in \mathcal{R} : P(f_{T|X}(t|X) < C) = 0\}$.

⁶When unconfoundedness is violated, we can use the control function approach in triangular simultaneous equations models by including in the covariates some estimated control variables using instrumental variables, as in Imbens and Newey (2009). The influence function that accounts for estimating the control variables as generated regressors has derived in Corollary 2 in Lee (2015). It may be important to include the adjustment terms for the estimated control variables to achieve local robustness. Using such an influence function to construct the corresponding DML estimator is left for future research.

⁴Belloni et al. (2017) show how cross-fitting weakens the entropy condition and hence the sparsity assumption on nuisance Lasso estimator. The benefit of cross-fitting is further investigated by Wager and Athey (2018) for heterogeneous causal effects, Newey and Robins (2018) for double cross-fitting, and Cattaneo and Jansson (2019) for cross-fitting bootstrap.

Define the product kernel as $K_h(T_i - t) \equiv \prod_{j=1}^{d_T} k((T_{ji} - t_j)/h)/h^{d_T}$, where T_{ji} is the j th component of T_i and the kernel function $k(\cdot)$ satisfies [Assumption 2.2](#). Denote the roughness of k as $R_k \equiv \int_{-\infty}^{\infty} k(u)^2 du$ and $\kappa \equiv \int_{-\infty}^{\infty} u^2 k(u) du$.

Assumption 2.2 (Kernel). The second-order symmetric kernel function $k(\cdot)$ is bounded differentiable, that is, $\int_{-\infty}^{\infty} k(u) du = 1$, $\int_{-\infty}^{\infty} uk(u) du = 0$, and $0 < \kappa < \infty$. For some finite positive constants $C, \bar{U}$, and for some $\nu > 1$, $|dk(u)/du| \leq C|u|^{-\nu}$ for $|u| > \bar{U}$.

[Assumption 2.2](#) is standard in nonparametric kernel estimation and holds for commonly used kernel functions, such as Epanechnikov and Gaussian. By [Assumptions 2.1–2.2](#) and the same reasoning for the binary treatment, it is straightforward to show the identification,

$$\beta_t \equiv \mathbb{E}[Y(t)] = \int_{\mathcal{X}} \mathbb{E}[Y|T = t, X] dF_X(X) = \mathbb{E}[\gamma(t, X)] \quad (2.2)$$

$$\begin{aligned} &= \lim_{h \rightarrow 0} \int_{\mathcal{Z}} \frac{K_h(T - t)Y}{f_{T|X}(t|X)} dF_Z(Z) \\ &= \lim_{h \rightarrow 0} \mathbb{E} \left[\frac{K_h(T - t)Y}{f_{T|X}(t|X)} \right] \end{aligned} \quad (2.3)$$

for $t \in \mathcal{T}$.⁷ The expression in (2.2) motivates the class of regression-based (or imputation) estimators, while (2.3) motivates the class of inverse probability weighting estimators; see Section S3.1 in the appendix for further discussion. Now we introduce the double debiased machine learning estimator:

STEP 1. (Cross-fitting) For some fixed $L \in \{2, \dots, n\}$, randomly partition the observation indices into L distinct groups I_ℓ , $\ell = 1, \dots, L$, such that the sample size of each group is the largest integer smaller than n/L . For each $\ell = 1, \dots, L$, the estimators $\hat{\gamma}_\ell(t, x)$ for $\gamma(t, x) \equiv \mathbb{E}[Y|T = t, X = x]$ and $\hat{f}_\ell(t|x)$ for $f_{T|X}(t|x)$ use observations not in I_ℓ and satisfy [Assumption 2.4](#).

STEP 2. (Double robustness) Define the double debiased ML (DML) estimator as

$$\hat{\beta}_t \equiv \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \left\{ \hat{\gamma}_\ell(t, X_i) + \frac{K_h(T_i - t)}{\hat{f}_\ell(t|X_i)} (Y_i - \hat{\gamma}_\ell(t, X_i)) \right\}. \quad (2.4)$$

STEP 3. (Partial effect) Let $t^+ \equiv (t_1 + \eta/2, t_2, \dots, t_{d_T})'$ and $t^- \equiv (t_1 - \eta/2, t_2, \dots, t_{d_T})'$, where η is a positive sequence converging to zero as $n \rightarrow \infty$. We estimate the partial effect of the first component of the continuous treatment $\theta_t \equiv \partial \beta_t / \partial t_1$ by $\hat{\theta}_t \equiv (\hat{\beta}_{t^+} - \hat{\beta}_{t^-}) / \eta$.

The number of folds in cross-fitting L is not random and typically small, such as five or ten in practice; see, for example, CCDDHNR. When there is no sample splitting ($L = 1$), $\hat{\gamma}_1$ and $\hat{f}_1$ use all observations in the full sample. Then the DML estimator $\hat{\beta}_t$ in (2.4) is the estimator in Su, Ura, and Zhang (2019) with Lasso-type nuisance function estimators.

Our inference theory requires the estimators $\hat{\gamma}_\ell$ and $\hat{f}_\ell$ in Step 1 to satisfy [Assumption 2.4](#). We define the *partial $L_2(tX)$ norm* for $t \in \mathcal{T}$ as $\|\hat{\gamma}_\ell - \gamma\|_{F_{tX}} \equiv \|\hat{\gamma}_\ell(t, X) - \gamma(t, X)\|_{F_{tX}} \equiv (\int_{\mathcal{X}} (\hat{\gamma}_\ell(t, x) - \gamma(t, x))^2 f_{TX}(t, x) dx)^{1/2}$ and $\|\hat{f}_\ell - f_{T|X}\|_{F_{tX}} \equiv \|\hat{f}_\ell(t|X) - f_{T|X}(t|X)\|_{F_{tX}} \equiv (\int_{\mathcal{X}} (\hat{f}_\ell(t|x) - f_{T|X}(t|x))^2 f_{TX}(t, x) dx)^{1/2}$, where the joint distribution $F_{TX}(t, X)$ is evaluated at a fixed value of T equal to t .

Assumption 2.3. There exist functions $\bar{\gamma}(t, x)$ and $\bar{f}(t, x)$ that are three-times differentiable w.r.t. t with all three derivatives being bounded uniformly over $\mathcal{T}_0 \times \mathcal{X}$ and $\inf_{t \in \mathcal{T}} \text{ess inf}_{x \in \mathcal{X}} \bar{f}(t, x) \geq c$ for some positive constant c .

Assumption 2.4. For $\ell = 1, \dots, L$ and $t \in \mathcal{T}$, (a) $\|\hat{\gamma}_\ell - \bar{\gamma}\|_{F_{tX}} = o_p(1)$ and $\|\hat{f}_\ell - \bar{f}\|_{F_{tX}} = o_p(1)$, (b) $\sqrt{nh^{d_T}} \|\hat{f}_\ell - f_{T|X}\|_{F_{tX}} \|\hat{\gamma}_\ell - \gamma\|_{F_{tX}} \xrightarrow{p} 0$; (c) Either $\bar{\gamma} = \gamma$ or $\bar{f} = f_{T|X}$.

The nuisance function estimators $\hat{\gamma}_\ell$ and $\hat{f}_\ell$ converge to some fixed functions $\bar{\gamma}$ and $\bar{f}$ in [Assumption 2.3](#) in the sense of [Assumption 2.4\(a\)\(b\)](#). [Assumption 2.4\(c\)](#) allows one of the nuisance functions to be misspecified, or requires at least one of the nuisance function estimators to be consistent. When both nuisance function estimators are nonparametric, there is no concern of misspecification as $\bar{\gamma} = \gamma$ and $\bar{f} = f_{T|X}$. Our doubly robust inference is useful when we estimate the nuisance functions by semiparametric methods or regularization methods, such as Lasso. [Assumptions 2.4\(b\)](#) and (c) imply that if one nuisance function is misspecified, then the other needs to be estimated consistently at a rate faster than $\sqrt{nh^{d_T}}$. To see why, if $\bar{\gamma} \neq \gamma$, then $\|\hat{\gamma}_\ell - \gamma\|_{F_{tX}} = O_p(1)$, and (b) requires $\sqrt{nh^{d_T}} \|\hat{f}_\ell - f_{T|X}\|_{F_{tX}} = o_p(1)$. So for the doubly robust inference to allow misspecification of one of the nuisance functions, we require both nuisance function estimators to converge at a rate faster than $\sqrt{nh^{d_T}}$. This rate condition holds for semiparametric methods with a regular $\sqrt{n}$ convergence rate. Su, Ura, and Zhang (2019) provide the convergence rates of a Lasso-type estimator. Kennedy et al. (2017) show such double robustness using a stronger uniform norm for a local linear estimation.

In [Section 3.1](#), we provide sufficient low-level conditions for [Assumption 2.4](#) when the nuisance function estimators are kernel estimators, series, and the deep neural networks in Farrell, Liang, and Misra (2021b). The partial L_2 convergence rate also appears in Kennedy et al. (2017), the conditional average treatment effect in Fan et al. (2022), and covariate adjustments in regression discontinuity designs in Noack, Olma, and Rothe (2021).

Remark 2.1. Consider the binary treatment effect in CCDDHNR as a heuristic example of a regular parameter that can be estimated at $\sqrt{n}$ -rate. When $\bar{\gamma} \neq \gamma$, [Assumption 2.4](#) suggests that the corresponding condition on the propensity score

⁷For identification, we only need $\inf_{t \in \mathcal{T}} \text{ess inf}_{x \in \mathcal{X}} f_{T|X}(t|x) > 0$ for (2.2) and additionally $\gamma(t, x)$ and $f_{T|X}(t, x)$ to be continuous in t for (2.3). Such standard identifying conditions are weaker than [Assumption 2.1](#) used for the inference theory. The common support (or strong overlap) [Assumption 2.1\(b\)](#) is standard in the continuous treatment literature (e.g., Hirano, Imbens, and Ridder 2003; Su, Ura, and Zhang 2019) and also in the binary treatment literature to guarantee $\sqrt{n}$ -consistent estimators and regular identification. We note that [Assumption 2.1\(b\)](#) should be made with care in practice and is strong especially with many control variables. A possible approach to relaxing [Assumption 2.1\(b\)](#) is to define the object of interest by a common support via fixed trimming, for example Lee (2018).

estimator $\hat{f}_\ell$ would be $\sqrt{n}\|\hat{f}_\ell - f_{T|X}\|_{F_{TX}} = o_p(1)$, which is not feasible. Therefore, we conjecture that the inference theory of the DML estimators for the regular estimands in the semiparametric models of CCDDHNR cannot allow for misspecification of the nuisance functions. In contrast, our DML estimator for the non-regular estimand allows for doubly robust inference. Tan (2020) develops doubly robust inference for the average binary treatment effect by carefully chosen loss functions to estimate the nuisance functions.⁸

Remark 2.2 (Kernel localization). We discuss the construction of the doubly robust moment function (1.1) by Gateaux derivative and a local Riesz representer; details are in Section S3 in the appendix. From the literature on estimating regular parameters, the Gateaux derivative is fundamental to construct estimators with desired properties, such as bias reduction and double robustness in Carone, Luedtke, and van der Laan (2018) and Ichimura and Newey (2022). The partial mean β_t is a marginal integration over the conditional distribution of Y given (T, X) and the marginal distribution of X , fixing the value of T at t . As a result, the Gateaux derivative and the Riesz representer depend on the choice of the distribution f_T^h that belongs to a family of distributions approaching a point mass at T as $h \rightarrow 0$. So the asymptotic distribution of $\hat{\beta}_t$ depends on the choice of f_T^h that is the kernel function $K_h(T - t)$. Moreover, to construct the DML estimator of a linear functional of β_t that preserves the good properties, the corresponding moment function is simply the linear functional of the moment function of β_t . To our best knowledge, this is the first explicit calculation of Gateaux derivative for such a non-regular nonparametric parameter. Alternatively Kennedy et al. (2017) construct a “pseudo-outcome” that is motivated from the doubly robust and efficient influence function of the regular semiparametric parameter $\int \beta_t f_T(t) dt$. Then they locally regress the pseudo-outcome on T at t using a kernel to estimate β_t . Semenova and Chernozhukov (2020) illustrate in an example to estimate β_t by the best linear projection of an “orthogonal signal of the outcome” which is the same “pseudo-outcome” proposed by Kennedy et al. (2017).

A second motivation of the moment function is adding to the influence function of the regression (or imputation) estimator $n^{-1} \sum_{i=1}^n \hat{\gamma}(t, X_i)$ the adjustment term from a kernel-based estimator $\hat{\gamma}$. A series estimator $\hat{\gamma}$ yields a different adjustment. These distinct features of continuous treatments are in contrast to the regular binary treatment case, where different nonparametric nuisance function estimators $\hat{\gamma}$ result in the same efficient influence function.

2.1. Conditional Density Estimation

We propose an estimator for the *reciprocal of the generalized propensity score* (GPS), that is, $1/f_{T|X}(t|x)$, when $d_T = 1$. The

estimator avoids plugging in a small estimate in the denominator, and the estimate is positive by construction. When $d_T > 1$, we propose an estimator for the GPS. We can use various nonparametric and ML methods designed for the conditional expectation. We provide a root-mean-square convergence rate. In Section 3.1, we demonstrate these generic GPS estimators using the deep neural networks in Farrell, Liang, and Misra (2021b) and show how Assumption 2.4 can be verified.

The theory of ML methods in estimating the conditional density is less developed compared with estimating the conditional expectation. Alternative estimators for estimating the GPS can be the kernel density estimator, the artificial neural networks in Chen and White (1999), the Lasso methods in Su, Ura, and Zhang (2019), Belloni, Chernozhukov, and Kato (2019), Klossin (2021),⁹ or the series cross-validated method in Zhang (2022).

It is known that for any CDF F , $\frac{d}{du} F^{-1}(u) = \frac{1}{F'(F^{-1}(u))}$ for $u \in (0, 1)$. So $\frac{1}{f_{T|X}(t|x)} = \frac{\partial}{\partial u} F_{T|X}^{-1}(u|x) \Big|_{u=F_{T|X}(t|x)}$. Inspired by the idea in Koenker (1994) and Belloni, Chernozhukov, and Kato (2019), we estimate $\frac{1}{f_{T|X}(t|x)}$ by a numerical differentiation estimator, labeled as ReGPS,

$$\widehat{\frac{1}{f_{T|X}(t|x)}} = \frac{\hat{F}_{T|X}^{-1}(\hat{F}_{T|X}(t|x) + \epsilon|x) - \hat{F}_{T|X}^{-1}(\hat{F}_{T|X}(t|x) - \epsilon|x)}{2\epsilon}, \quad (\text{ReGPS})$$

where $\epsilon = \epsilon_n$ is a positive sequence vanishing as n grows and $\hat{F}_{T|X}(t|x) \pm \epsilon \in (0, 1)$. By a standard algebra, the conditional CDF $F_{T|X}(t|x) = \lim_{h_1 \rightarrow 0} \mathbb{E} \left[\Phi \left(\frac{t-T}{h_1} \right) \Big| X = x \right]$, where Φ is the CDF of a standard normal random variable and $h_1 = h_{1n}$ is a bandwidth sequence vanishing as n grows. Let $\hat{\mu}(W; x)$ be a generic estimator of the conditional expectation $\mathbb{E}[W|X = x]$ for an outcome variable W . Then we estimate $F_{T|X}(t|x)$ by $\hat{F}_{T|X}(t|x) = \hat{\mu} \left(\Phi \left(\frac{t-T}{h_1} \right); x \right)$ with a transformed outcome variable of T , $W = \Phi \left(\frac{t-T}{h_1} \right)$. The conditional u -quantile function $F_{T|X}^{-1}(u|x)$ is estimated by the generalized inverse function $\hat{F}_{T|X}^{-1}(u|x) = \inf_{t \in \mathcal{T}} \{t : \hat{F}_{T|X}(t|x) \geq u\}$. When $\hat{F}_{T|X}(t|x)$ is continuous in t , $\hat{F}_{T|X}^{-1}(u|x)$ is strictly increasing. Then the resulting estimator $1/\widehat{f_{T|X}(t|x)} > 0$.

Denote the standard root-mean-square norm, or the $L_2(X)$ norm, of a random vector X with distribution F_X as $\|\hat{\mu}(W; X) - \mathbb{E}[W|X]\|_{F_X} \equiv \left(\int_{\mathcal{X}} (\hat{\mu}(W; x) - \mathbb{E}[W|X = x])^2 f_X(x) dx \right)^{1/2}$ for a random variable W .

Lemma 2.1 (ReGPS). Let an estimator of $\mathbb{E} \left[\Phi \left(\frac{t-T}{h_1} \right) \Big| X = x \right]$, denoted as $\hat{\mu} \left(\Phi \left(\frac{t-T}{h_1} \right); x \right)$, be continuous in $t \in \mathcal{T}$ and satisfy $\sup_{t \in \mathcal{T}} \left\| \hat{\mu} \left(\Phi \left(\frac{t-T}{h_1} \right); X \right) - \mathbb{E} \left[\Phi \left(\frac{t-T}{h_1} \right) \Big| X \right] \right\|_{F_X} = O_p(R_{1n})$ for a sequence of constants R_{1n} . Assume $F_{T|X}$ to be two-times differentiable w.r.t. t with all two derivatives being bounded

⁸For the semiparametric GMM, Chernozhukov et al. (2022) consider *estimator-based* orthogonalization, rather than *model-based*, in the sense that their parameter of interest is defined by the probability limit of the first-step estimator. Specifically in our setting for example, the parameter of interest in CEINR would be defined as $\beta_t = \mathbb{E}[\tilde{\gamma}(t, X)]$, which is not our model-based causal parameter $\beta_t = \mathbb{E}[Y(t)]$ if $\tilde{\gamma}(t, x) \neq \mathbb{E}[Y|T = t, X = x]$. Then Chernozhukov et al. (2022) provide the standard errors robust to such misspecification.

⁹Su, Ura, and Zhang (2019) estimate the GPS by the numerical derivative of the conditional CDF of T given X , which is estimated by the logistic distributional Lasso regression in Belloni et al. (2017). Belloni, Chernozhukov, and Kato (2019) use a ℓ_1 -penalized quantile regression to estimate the conditional density function. They use the density estimates as weights in the numerator in the weighted Lasso estimation, different from the GPS in the denominator in our case.

uniformly over $\mathcal{T} \times \mathcal{X}$. Then the ReGPS estimator $\widehat{1/f_{T|X}}$ satisfies $\sup_{t \in \mathcal{T}} \|\widehat{1/f_{T|X}}(t|X) - 1/f_{T|X}(t|X)\|_{F_X} = O_p(R_{1n}\epsilon^{-1} + h_1^2\epsilon^{-1} + \epsilon^2)$ and $\sup_{t \in \mathcal{T}} \|\widehat{1/f_{T|X}} - 1/f_{T|X}\|_{F_{tX}} = O_p(R_{1n}\epsilon^{-1} + h_1^2\epsilon^{-1} + \epsilon^2)$.

Assumption 2.4 specifies the root-mean-square convergence rate of our conditional density estimator. Thus, as long as the root-mean-square convergence rate of a ML method (R_{1n}) for the conditional expectation function is available, **Assumption 2.4** can be satisfied with a suitable bandwidth h_1 and ϵ . Then we are able to use such a ML method to estimate the conditional density, as illustrated in [Section 3.1](#).

Note that $\hat{F}_{T|X}^{-1}(u|x)$ estimates the conditional quantile function. The ReGPS estimator inverts the CDF estimate and allows to apply various ML methods for conditional expectation functions. Klossin (2021) and Chernozhukov, Newey, and Singh (2022b) also avoid inverting estimated propensity scores for average treatment effects by directly estimating the reciprocal of the GPS or Riesz representer. Their regularization method (Lasso or generalized Dantzig selector) is not nonparametric and assumes approximate sparsity. ReGPS allows various nonparametric and ML methods which satisfy the high-level rate conditions, including regularization methods like Lasso.

When $d_T > 1$, we estimate the conditional density function $f_{T|X}(t|x)$ by

$$\hat{f}_{T|X}(t|x) = h_1^{-d_T} \hat{\mu}(h_1^{d_T} g_{h_1}(T - t); x), \quad (\text{MultiGPS})$$

labeled as MultiGPS, where the bandwidth h_1 is a positive sequence vanishing as n grows, the product kernel $g_{h_1}(T_i - t) \equiv \Pi_{j=1}^{d_T} g((T_{ji} - t_j)/h_1)/h_1^{d_T}$. We can choose $g(\cdot)$ to be the Gaussian kernel.

Lemma 2.2 (MultiGPS). Let an estimator of $\mathbb{E}[h_1^{d_T} g_{h_1}(T - t)|X = x]$, denoted as $\hat{\mu}(h_1^{d_T} g_{h_1}(T - t); x)$, satisfy $\|\hat{\mu}(h_1^{d_T} g_{h_1}(T - t); X) - \mathbb{E}[h_1^{d_T} g_{h_1}(T - t)|X]\|_{F_X} = O_p(R_{1n})$ for a sequence of constants R_{1n} and $t \in \mathcal{T}$. Let $g(\cdot)$ satisfy **Assumption 2.2** with $g(\cdot)$ in place of $k(\cdot)$ and with an unbounded support. Assume $f_{T|X}(t|x)$ to be two-times differentiable w.r.t. t with all two derivatives being bounded uniformly over $\mathcal{T} \times \mathcal{X}$. Then the MultiGPS estimator $\hat{f}_{T|X}$ satisfies $\|\hat{f}_{T|X}(t|X) - f_{T|X}(t|X)\|_{F_X} = O_p(R_{1n}h_1^{-d_T} + h_1^2)$ and $\|\hat{f}_{T|X} - f_{T|X}\|_{F_{tX}} = O_p(R_{1n}h_1^{-d_T} + h_1^2)$ for $t \in \mathcal{T}$. Further assume $\sup_{t \in \mathcal{T}} \|\hat{\mu}(h_1^{d_T} g_{h_1}(T - t); X) - \mathbb{E}[h_1^{d_T} g_{h_1}(T - t)|X]\|_{F_X} = O_p(R_{1n})$ for a constant sequence R_{1n} . Then $\sup_{t \in \mathcal{T}} \|\hat{f}_{T|X}(t|X) - f_{T|X}(t|X)\|_{F_X} = O_p(R_{1n}h_1^{-d_T} + h_1^2)$ and $\sup_{t \in \mathcal{T}} \|\hat{f}_{T|X} - f_{T|X}\|_{F_{tX}} = O_p(R_{1n}h_1^{-d_T} + h_1^2)$.

3. Asymptotic Theory

We present the asymptotically linear representation and normality. [Section 3.1](#) provides low-level conditions for estimating the nuisance functions by the deep neural networks in Farrell, Liang, and Misra (2021b). [Section 3.2](#) provides sufficient high-level rate conditions using the standard L_2 norm.

Let $\partial_t^\nu g(t, \cdot) \equiv \partial^\nu g(t, \cdot)/\partial t^\nu$ denote the ν th partial derivative of a generic function g w.r.t. t , and $\partial_t \equiv \partial_t^1$.

Theorem 3.1 (Asymptotic normality). Let **Assumptions 2.1–2.4** hold. Let $h \rightarrow 0$, $nh^{d_T} \rightarrow \infty$, and $nh^{d_T+4} \rightarrow C \in [0, \infty)$. Then for $t \in \mathcal{T}$,

$$\sqrt{nh^{d_T}} (\hat{\beta}_t - \beta_t) = \sqrt{\frac{h^{d_T}}{n}} \sum_{i=1}^n \left\{ \frac{K_h(T_i - t)}{\hat{f}(t, X_i)} (Y_i - \bar{\gamma}(t, X_i)) + \bar{\gamma}(t, X_i) - \beta_t \right\} + o_p(1). \quad (3.5)$$

Further let $\mathbb{E}[|Y - \bar{\gamma}(T, X)|^3|T = t, X]$ and its derivatives w.r.t. t be bounded uniformly over $\mathcal{T} \times \mathcal{X}$. Let $\int_{-\infty}^{\infty} k(u)^3 du < \infty$. Then $\sqrt{nh^{d_T}} (\hat{\beta}_t - \beta_t - h^2 \mathbf{B}_t) \xrightarrow{d} \mathcal{N}(0, \mathbf{V}_t)$, where $\mathbf{B}_t \equiv \sum_{j=1}^{d_T} \mathbb{E} \left[\left(2\partial_{t_j} \bar{\gamma}(t, X) \partial_{t_j} f_{T|X}(t|X) + \partial_{t_j}^2 \bar{\gamma}(t, X) f_{T|X}(t|X) + (\gamma(t, x) - \bar{\gamma}(t, x)) \partial_{t_j}^2 \gamma(t, X) \right) / \bar{f}(t, X) \right] \kappa/2$ and $\mathbf{V}_t \equiv \mathbb{E}[\mathbb{E}[(Y - \bar{\gamma}(t, X))^2|T = t, X] f_{T|X}(t, X) / \bar{f}(t, X)^2] R_k^{d_T}$.

The second part in the influence function¹⁰ in (3.5) $n^{-1} \sum_{i=1}^n \bar{\gamma}(t, X_i) - \beta_t = O_p(1/\sqrt{n}) = o_p(1/\sqrt{nh^{d_T}})$ and hence does not contribute to the first-order asymptotic variance $\mathbf{V}_t$. We keep these smaller-order terms to show that the nuisance function estimators have no first-order influence on the asymptotic distribution.

Theorem 3.1 is fundamental for inference, such as constructing confidence intervals and the optimal bandwidth h that minimizes the asymptotic mean squared error. We propose an estimator for the leading bias $\mathbf{B}_t$, inspired by the idea in Powell and Stoker (1996). Let the notation $\hat{\beta}_t = \hat{\beta}_{t,b}$ be explicit on the bandwidth b and $\hat{\mathbf{B}}_t \equiv (\hat{\beta}_{t,b} - \hat{\beta}_{t,ab})/(b^2(1 - a^2))$ with a pre-specified fixed scaling parameter $a \in (0, 1)$.

We can estimate the asymptotic variance $\mathbf{V}_t$ by the sample variance of the estimated influence function $\hat{\mathbf{V}}_t \equiv h^{d_T} n^{-1} \sum_{\ell=1}^L \sum_{i \in I_\ell} \hat{\psi}_{i\ell}^2$, where $\hat{\psi}_{i\ell} \equiv K_h(T_i - t)(Y_i - \hat{\gamma}_\ell(t, X_i))/\hat{f}_\ell(t|X_i) + \hat{\gamma}_\ell(t, X_i) - \hat{\beta}_t$. Then we propose a data-driven bandwidth $\hat{h}_t \equiv (d_T \hat{\mathbf{V}}_t / (4\hat{\mathbf{B}}_t^2))^{1/(d_T+4)} n^{-1/(d_T+4)}$ to consistently estimate the optimal bandwidth that minimizes the asymptotic mean squared error (AMSE) given in [Theorem 3.2](#).

Assumption 3.1. For $\ell = 1, \dots, L$ and $t \in \mathcal{T}$, (a) $\|(\hat{\gamma}_\ell - \bar{\gamma})(\hat{f}_\ell - \bar{f})\|_{F_{tX}} = o_p(1)$, (b) $\|(\hat{\gamma}_\ell - \bar{\gamma})^2(\hat{f}_\ell - \bar{f})^2\|_{F_{tX}} = O_p(1)$, $\|(\hat{\gamma}_\ell - \bar{\gamma})^2\|_{F_{tX}} = O_p(1)$, and $\|(\hat{f}_\ell - \bar{f})^2\|_{F_{tX}} = O_p(1)$; (c) $\mathbb{E}[(Y - \bar{\gamma}(t, x))^4|T = t, X = x]$ and its derivatives w.r.t. t are bounded uniformly over $\mathcal{T} \times \mathcal{X}$ and $\int_{-\infty}^{\infty} k(u)^4 du < \infty$; (d) $b \rightarrow 0$ and $nb^{d_T+4} \rightarrow \infty$. $\int_{-\infty}^{\infty} k(u)k(u/a)du < \infty$ for $a \in (0, 1)$.

Theorem 3.2 (AMSE optimal bandwidth for $\hat{\beta}_t$). Let the conditions in [Theorem 3.1](#) hold. For $t \in \mathcal{T}$, if $\mathbf{B}_t$ is nonzero, then the bandwidth that minimizes the asymptotic mean squared error is $h_t^* = (d_T \mathbf{V}_t / (4\mathbf{B}_t^2))^{1/(d_T+4)} n^{-1/(d_T+4)}$. Further let **Assump-**

¹⁰For our non-regular parameters, we borrow the terminology ‘‘influence function’’ in estimating a regular parameter that is $\sqrt{n}$ -estimable. An influence function gives the first-order asymptotic effect of a single observation on the estimator. The estimator is asymptotically equivalent to a sample average of the influence function. See Hampel (1974) and Ichimura and Newey (2022), for example.

tion 3.1 hold. Then $\hat{V}_t - V_t = o_p(1)$, $\hat{B}_t - B_t = o_p(1)$, and $\hat{h}_t/h_t^* - 1 = o_p(1)$.

Assumption 3.1(a)–(c) are for the consistency of $\hat{V}_t$, and (d) is for $\hat{B}_t$. The condition (a) strengthens **Assumption 2.4**(a), and (b) is mild boundedness conditions that are implied when $\hat{\gamma}_\ell$ and $\hat{f}_\ell$ are bounded uniformly. In practice, we may use different Step 1 estimators in $\hat{V}_t$ and $\hat{B}_t$ due to different high-level conditions.

A common approach is to choose an undersmoothing bandwidth h smaller than h_t^* such that the bias is first-order asymptotically negligible, that is, $h^2\sqrt{nh^{d_T}} \rightarrow 0$. Then we can construct the usual $(1 - \alpha) \times 100\%$ point-wise confidence interval $\left[\hat{\beta}_t \pm \Phi^{-1}(1 - \alpha/2)\sqrt{\hat{V}_t/(nh^{d_T})}\right]$, where Φ is the CDF of $\mathcal{N}(0, 1)$. Alternatively, we may consider a bias-corrected estimator $\hat{\beta}_t - h^2\hat{B}_t$. Calonico, Cattaneo, and Farrell (2018) show that the AMSE optimal bandwidth of the original estimator ($\hat{h}_t$) is feasible in different contexts. Takatsu and Westling (2024) develop such robust bias-corrected inference for the local linear estimator in Kennedy et al. (2017).

Theorem 3.3 (Partial effect). Let the conditions in **Theorem 3.1** hold. Assume that $f_Z(y, t, x)$ is four-times differentiable w.r.t. t with all four derivatives being bounded uniformly over $(y, t', x') \in \mathcal{Z}$. Let $\eta/h \rightarrow 0$, $nh^{d_T+2} \rightarrow \infty$, $nh^{d_T+2}\eta^2 \rightarrow 0$, and $nh^{d_T+6} \rightarrow C \in [0, \infty)$. Let $\int_{-\infty}^{\infty} k'''(u)^2 du < \infty$. Assume that for each $\ell = 1, \dots, L$ and for any $t \in \mathcal{T}$, (a) $\eta^{-1}h\|\hat{\gamma}_\ell - \bar{\gamma}\|_{F_{IX}} = o_p(1)$ and $\eta^{-1}h\|\hat{f}_\ell - \bar{f}\|_{F_{IX}} = o_p(1)$; (b) $\eta^{-1}h\sqrt{nh^{d_T}}\|\hat{f}_\ell - f_{T|X}\|_{F_{IX}}\|\hat{\gamma}_\ell - \gamma\|_{F_{IX}} \xrightarrow{p} 0$. Then for any $t \in \mathcal{T}$,

$$\begin{aligned} & \sqrt{nh^{d_T+2}}(\hat{\theta}_t - \theta_t) \\ &= \sqrt{\frac{h^{d_T+2}}{n}} \sum_{i=1}^n \frac{\partial}{\partial t_1} K_h(T_i - t) \frac{Y_i - \bar{\gamma}(t, X_i)}{\bar{f}(t, X_i)} + o_p(1) \quad (3.6) \end{aligned}$$

and $\sqrt{nh^{d_T+2}}(\hat{\theta}_t - \theta_t - h^2B_t^\theta) \xrightarrow{d} \mathcal{N}(0, V_t^\theta)$, where $V_t^\theta \equiv V_t \int k'(u)^2 du R_k^{-1}$ with V_t given in **Theorem 3.1** and $B_t^\theta \equiv B_{\theta_t} + B_{\theta_t}^{mis}$. The exact formulas of B_{θ_t} and $B_{\theta_t}^{mis}$ are given in equations (S.22) and (S.23), respectively in the appendix.

The conditions (a) and (b) in **Theorem 3.3** strengthen **Assumption 2.4** for $\hat{\beta}_t$ and imply that η cannot be too small and depends on the precision of the nuisance function estimators. The bias $B_{\theta_t}^{mis}$ is due to misspecifying one of the nuisance functions and is zero when both nuisance functions are correctly specified or estimated by nonparametric methods. Following the same approach for $\hat{\beta}_t$, we provide a data-driven bandwidth for $\hat{\theta}_t$ and its asymptotic theory for $\hat{\theta}_t$ in Section S1 in the appendix.

3.1. Nuisance Function Estimators

We show that the high-level conditions on the convergence rates in **Assumption 2.4** are attainable by the nonparametric and ML methods: kernel, series, and the deep neural networks in Farrell, Liang, and Misra (2021b), where the dimension of the control variables d_X is fixed. Lasso methods have been extensively

studied in Su, Ura, and Zhang (2019), Sasaki and Ura (2023), and Sasaki, Ura, and Zhang (2022), where d_X can grow with n . These ML methods require different low-level conditions, such as dimensionality, smoothness, and tuning parameters.

Consider the conditional density estimator MultiGPS $\hat{f}_{T|X}$ given in **Section 2.1** for example. By **Lemma 2.2**, **Assumption 2.4**(b) requires $\sqrt{nh^{d_T}}\|\hat{f} - f_{T|X}\|_{F_{IX}}\|\hat{\gamma} - \gamma\|_{F_{IX}} = \sqrt{nh^{d_T}}(R_{1n}h_1^{-d_T} + h_1^2)\|\hat{\gamma} - \gamma\|_{F_{IX}} = o_p(1)$. Therefore, we need to obtain the partial $L_2(tX)$ convergence rate $\|\hat{\gamma} - \gamma\|_{F_{IX}}$ and the standard $L_2(X)$ convergence rate R_{1n} .

We seek theoretical results, such as the above rate conditions, for insights on selection of the tuning parameters in practice that is challenging and under-developed in the ML literature. We may use the optimal choices for the nuisance function estimators, as they do not affect the first-order asymptotics. A common method is cross-validation. We may choose the optimal rates for the bandwidths for $\hat{\gamma}$ and $\hat{\mu}(h_1^{d_T}g_{h_1}(T - t); x)$ that respectively minimize $\|\hat{\gamma} - \gamma\|_{F_{IX}}$ and $\|\hat{\mu}(h_1^{d_T}g_{h_1}(T - t); X) - \mathbb{E}[h_1^{d_T}g_{h_1}(T - t)|X]\|_{F_X}$, which might be available in the literature. Similarly we can derive the optimal $h_1^* \propto R_{1n}^{1/(2+d_T)}$.

Next we propose a deep MLP-ReLU network kernel estimator for $\gamma(t, x)$ (labeled as Kernel NN) and derive its $L_2(tX)$ convergence rate. We illustrate the low-level conditions of conventional kernel and series estimators in Section S4 in the appendix, as the calculations are rather standard. These results on the $L_2(tX)$ convergence rates for neural networks and series are new and nontrivial extensions of existing results in the literature.

We consider the deep neural networks in Farrell, Liang, and Misra (2021b) (FLM, hereafter) that use the fully connected feedforward neural networks (multilayer perceptron, or MLP) and the nonsmooth rectified linear units (ReLU) activation function. We propose a deep MLP-ReLU network kernel estimator for $\gamma(t, x)$. The proposed estimator serves the purpose to conveniently apply the $L_2(TX)$ convergence rate given in FLM to obtain the $L_2(tX)$ convergence rate. So we can deliver valid asymptotic inference for β_t and θ_t following deep learning. In this section, we closely follow the notations in FLM for easy reference, by slightly abusing our notations.

We define the *deep MLP-ReLU network kernel estimator* for $t \in \mathcal{T}$ as

$$\hat{f}_{tb} \in \arg \min_{f_\theta \in \mathcal{F}_{MLP}, \|f_\theta\|_\infty \leq 2M} \sum_{i=1}^n \frac{1}{2} (Y - f(X))^2 K_b(T - t), \quad (3.7)$$

where $\mathcal{F}_{MLP}$ is the MLP class, M is an absolute constant, and θ depending on t collects the weights and constants over all nodes. We consider a kernel-weighted loss function for $t \in \mathcal{T}$ in (3.7), where a product kernel $K_b(T - t) \equiv \prod_{j=1}^{d_T} k((T_j - t_j)/b)/b^{d_T}$ with a kernel function k and a positive sequence of bandwidth $b = b_n$ vanishing as n grows. We refer the details of the MLP-ReLU network estimators to FLM. Then we obtain $\hat{\gamma}(t, x) = \hat{f}_{tb}(x)$.

Denote the derivative of a function $f(x)$ as $D_x^\alpha f(x) = \frac{\partial^{|\alpha|} f(x)}{\partial x_1^{\alpha_1} \dots \partial x_{d_X}^{\alpha_{d_X}}}$, where $\alpha = (\alpha_1, \dots, \alpha_{d_X})$ and $|\alpha| = \alpha_1 + \dots + \alpha_{d_X}$.

Assumption 3.2 (DNN). (a) For $t \in \mathcal{T}$, the second derivatives of $f_{T|X}(t, x)$ and $\gamma(t, x)$ w.r.t. t are bounded and continuous

uniformly over $x \in \mathcal{X}$; (b) X are continuously distributed with support $\mathcal{X} = [-1, 1]^{d_X}$ for fixed d_X ,¹¹ and for an absolute constant $M > 0$, $\mathcal{Y} \subset [-M, M]$; (c) For $r \in \mathcal{N}_+$ and some finite positive constant c , $\max_{\alpha, |\alpha| \leq r} \text{ess sup}_{y \in \mathcal{Y}, t \in \mathcal{T}, x \in \mathcal{X}} |\mathcal{D}_x^\alpha f_{Y|X}(y, t|x)| \leq c$; (d) $\|f_{T|X}(t|\cdot)\|_\infty \leq c$ for some finite positive constant c .

Assumption 3.2(a) is due to the kernel weight $K_b(T - t)$ in the loss function. **Assumption 3.2(b)–(d)** collect assumptions for applying Theorem 1 in FLM. Detailed discussion on these assumptions is referred to FLM.

Theorem 3.4 (DNN). Let $\hat{f}_{tb}$ be the deep MLP-ReLU network kernel estimator defined by (3.7). Let **Assumption 3.2** hold. Let $k(\cdot)$ satisfy **Assumption 2.2** with $k(\cdot)$ in place of $k(\cdot)$ and with a bounded support. Let width $H \asymp (nb^{2d_T})^{\frac{d_X}{2(r+d_X)}} \log^2(nb^{2d_T})$ and depth $L \asymp \log(nb^{2d_T})$. Then for $t \in \mathcal{T}$, $\|\hat{f}_{tb} - \gamma\|_{F_{TX}}^2 \leq C \cdot \left\{ (nb^{2d_T})^{-\frac{r}{r+d_X}} \log^8 n + \log \log n / (nb^{d_T}) + b^2 \right\}$ with probability approaching one as $n \rightarrow \infty$, for a constant $C > 0$ independent of n , which may depend on d_X , M , and other fixed constants.

We can apply deep neural networks to the GPS estimation proposed in Section 2.1. The ReGPS estimator can use $\hat{\mu}(\Phi((t - T)/h_1); x) = \hat{f}_{MLP-ReGPS}(x)$ the MLP estimator in FLM with the unweighted loss function:

$$\hat{f}_{MLP-ReGPS} \in \arg \min_{f_\theta \in \mathcal{F}_{MLP}, \|f_\theta\|_\infty \leq 2M} \sum_{i=1}^n (\Phi((t - T_i)/h_1) - f_\theta(X_i))^2. \quad (3.8)$$

The MultiGPS estimator can use $\hat{\mu}(h_1^{d_T} g_{h_1}(T - t); x) = \hat{f}_{MLP-MultiGPS}(x)$:

$$\hat{f}_{MLP-MultiGPS} \in \arg \min_{f_\theta \in \mathcal{F}_{MLP}, \|f_\theta\|_\infty \leq 2M} \sum_{i=1}^n (h_1^{d_T} g_{h_1}(T_i - t) - f_\theta(X_i))^2. \quad (3.9)$$

Lemma 3.1 (Theorem 1 in FLM). Let X be continuously distributed with support $\mathcal{X} = [-1, 1]^{d_X}$ for fixed d_X . Let width $H \asymp n^{\frac{d_X}{2(r+d_X)}} \log^2 n$ and depth $L \asymp \log n$. Let $R_{1n}^2 = n^{-\frac{r}{r+d_X}} \log^8 n + \log \log n / n$.

- (a) For $\hat{f}_{MLP-ReGPS}$ defined by (3.8), assume (A) $\max_{\alpha, |\alpha| \leq r} \text{ess sup}_{t \in \mathcal{T}, x \in [-1, 1]^{d_X}} |\mathcal{D}_x^\alpha F_{T|X}(t|x)| \leq c$ for some finite positive constant c . Let $\mathcal{T}_0 = [\underline{T}, \bar{T}] \subset \mathcal{R}$. Then $\sup_{t \in \mathcal{T}} \|\hat{f}_{MLP-ReGPS}(X) - \mathbb{E}[\Phi((t - T)/h_1)|X]\|_{F_X} = O_p(R_{1n})$.
- (b) For $\hat{f}_{MLP-MultiGPS}$ defined by (3.9), assume (B) $\max_{\alpha, |\alpha| \leq r} \text{ess sup}_{t \in \mathcal{T}, x \in [-1, 1]^{d_X}} |\mathcal{D}_x^\alpha f_{T|X}(t|x)| \leq c$ for some finite

positive constant c . Let $g(\cdot)$ satisfy **Assumption 2.2** in place of $k(\cdot)$ and with an unbounded support. Then $\sup_{t \in \mathcal{T}} \|\hat{f}_{MLP-MultiGPS}(X) - \mathbb{E}[h_1^{d_T} g_{h_1}(t - T)|X]\|_{F_X} = O_p(R_{1n})$.

We are ready to show that **Assumption 2.4** is attainable by the MLP-ReLU network estimators. Take the MultiGPS estimator in (3.9) with $d_T = 1$ for a simple example. **Assumption 2.4(b)** is $nh \left((nb^2)^{-\frac{r}{r+d_X}} \log^8 n + \log \log n / (nb) + b^2 \right) \left(h_1^{-2} (n^{-\frac{r}{r+d_X}} \log^8 n + \log \log n / n) + h_1^4 \right) \rightarrow 0$ by **Theorem 3.4**, **Lemmas 2.2**, and **3.1**. **Assumption 2.4(a)** is implied by $nb^2 \rightarrow \infty$ and $n^{r/(r+d_X)} h_1^2 \rightarrow \infty$. When $h = h_1 = b$, **Assumption 2.4(a)** holds by letting $n^{r/(r+d_X)} h^2 \rightarrow \infty$, and (b) holds by letting smoothness $r > d_X$. FLM discuss the same condition $r > d_X$ for the average treatment effect of a binary treatment variable. Specifically, FLM provide the corresponding $L_2(TX)$ convergence rate $\|\hat{\gamma}_\ell - \gamma\|_{F_{TX}}$. Similarly we note that this condition is not minimal but is sufficient to justify the practical use of the MLP-ReLU network estimators for valid inference on β_t and θ_t by our DML approach.

3.2. Standard L_2 Norm Rate

We provide sufficient rate conditions using the standard $L_2(TX)$ norm in **Assumption 3.3** as an alternative to the partial $L_2(tX)$ norm in **Assumption 2.4**. Let the $L_2(TX)$ norm be $\|\hat{\gamma}_\ell - \gamma\|_{F_{TX}} \equiv (\int_{\mathcal{T}} \int_{\mathcal{X}} (\hat{\gamma}_\ell(t, x) - \gamma(t, x))^2 f_{TX}(t, x) dx dt)^{1/2}$. We do not need to modify the rate condition on $\hat{f}_\ell$, as it equivalently uses the standard $L_2(X)$ norm for a given t . The cost of the more commonly used $L_2(TX)$ norm is losing the doubly robust inference and stricter regularity conditions, as discussed below.

Assumption 3.3. For $\ell = 1, \dots, L$ and $t \in \mathcal{T}$, (a) $h^{-d_T/2} \|\hat{\gamma}_\ell - \gamma\|_{F_{TX}} = o_p(1)$ and $\|\hat{f}_\ell - \bar{f}\|_{F_{TX}} = o_p(1)$; (b) $\sqrt{n} \|\hat{f}_\ell - \bar{f}\|_{F_{TX}} \|\hat{\gamma}_\ell - \gamma\|_{F_{TX}} = o_p(1)$; (c) $nh^{d_T+2} \rightarrow 0$. (d) $\hat{\gamma}_\ell$ is twice differentiable with respect to t with two derivatives being bounded uniformly over $\mathcal{T}_0 \times \mathcal{X}$.

Corollary 3.1 (Standard L_2 norm). Let the conditions in **Theorem 3.1** hold. Let **Assumption 3.3** instead of **Assumption 2.4** hold. Then for $t \in \mathcal{T}$, $\sqrt{nh^{d_T}}(\hat{\beta}_t - \beta_t) = \sqrt{h^{d_T}/n} \sum_{i=1}^n \{ (Y_i - \gamma(t, X_i)) K_h(T_i - t) / \bar{f}(t, X_i) + \gamma(t, X_i) - \beta_t \} + o_p(1) \xrightarrow{d} \mathcal{N}(0, V_t)$, where V_t is given in **Theorem 3.1** with $\bar{\gamma} = \gamma$.

Assumption 3.3(b) is the same rate condition for the regular semiparametric models in CCDDHNR, for example, $\hat{f}_{T|X}(t|x)$ is replaced with the propensity score $P(T = t|X = x)$ for a discrete treatment. We discuss the cost and implications of the high-level rate based on the standard L_2 norm. First, the inference theory cannot allow γ to be misspecified, while $f_{T|X}$ can be misspecified. Second, the convergence rates are faster than those required in **Assumption 2.4**. Heuristically the rate condition (a) uses the bounded kernel, resulting in a penalty $h^{-d_T/2}$ in the loose bound, as in Zimmert and Lechner (2019). Third, the choice of the bandwidth h is more restrictive. **Assumption 3.3(c)**

¹¹To simplify exposition, X is assumed to be continuous, but discrete variables with a fixed number of supporting points are allowed. Theorem 1 in Farrell, Liang, and Misra (2021a) find that the rate only depends on the dimension of the continuously distributed components.

implies undersmoothing, that is, the leading bias of $\hat{\beta}_t$ is first-order ignorable by $h^2\sqrt{nh^{d_T}} \rightarrow 0$.

For a specific example, consider $\hat{\gamma}$ to be FLM's estimator for the conditional expectation function $\mathbb{E}[Y|T, X]$. FLM provide the $L_2(TX)$ rate $\|\hat{\gamma}_\ell - \gamma\|_{F_{TX}}^2 = O_p\left(n^{-\frac{r}{r+d_X+d_T}} \log^8 n + \log \log n/n\right)$ with the smoothness r defined in [Assumption 3.2\(c\)](#). We need $r > d_T + d_X$ that is stronger than $r > d_X$, as discussed in [Section 3.1](#). Moreover comparing the rate conditions in [Assumption 2.4\(b\)](#) and [Assumption 3.3\(b\)](#) for $d_T = 1$, we can show that using the partial $L_2(tX)$ norm results in a tighter bound by $\sqrt{nh}\|\hat{\gamma}_\ell - \gamma\|_{F_{tX}} = o_p(\sqrt{n}\|\hat{\gamma}_\ell - \gamma\|_{F_{TX}})$ with $r = d_X + 1$, $b = h \propto n^{-a}$ and $a < 1/2$.

4. Simulated DML Estimator

We introduce Simulated DML estimator $\check{\beta}_t$ that enables the high-level rate conditions based on the standard $L_2(TX)$ norm, rather than the partial $L_2(tX)$ norm, and also permits the doubly robust inference. Following the estimation procedure given in [Section 2](#), Step 1 computes the nuisance function estimators $\hat{\gamma}_\ell$ and $\hat{f}_\ell$. In Step 2, let $U_i \equiv T_i h_0 + t$ where $\{T_i\}_{i \in I_\ell}$ are iid draws from $\{T_i\}_{i \in I_\ell}$ with replacement, for $i \in I_\ell$, $\ell \in \{1, \dots, L\}$, and h_0 is a positive sequence converging to zero as $n \rightarrow \infty$. The Simulated DML (SDML) estimator is defined as

$$\check{\beta}_t \equiv \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \left\{ \hat{\gamma}_\ell(U_i, X_i) + \frac{K_h(T_i - t)}{\hat{f}_\ell(t|X_i)} (Y_i - \hat{\gamma}_\ell(U_i, X_i)) \right\}.$$

The corresponding partial effect estimator $\check{\theta}_t \equiv (\check{\beta}_t^+ - \check{\beta}_t^-)/\eta$ as in [Step 3](#).

To get intuition, the SDML estimator $\check{\beta}_t$ uses $\hat{\gamma}_\ell(U_i, X_i)$ rather than $\hat{\gamma}_\ell(t, X_i)$ as in $\hat{\beta}_t$ that fixes the treatment value at the target t . The simulated U_i localizes the realized treatment values around t . Introducing such local variation enables the standard L_2 rate of $\hat{\gamma}$. Specifically T_i defined above follows the empirical distribution function of $\{T_i\}_{i \in I_\ell}$, denoted as $\hat{F}_{T_\ell}$. Therefore, U_i follows a CDF conditional on the sample $\{Z_i = (Y_i, X_i, T_i)\}_{i=1}^n$, $P(U_i \leq u | \{Z_i\}_{i=1}^n) = P(T_i h_0 + t \leq u | \{Z_i\}_{i=1}^n) = \hat{F}_{T_\ell}((u - t)/h_0)$. By a Taylor expansion, $\mathbb{E}[\gamma(U_i, X_i) | X_i] = \int_{\mathcal{T}} \gamma(t + sh_0, X_i) f_T(s) ds = \gamma(t, X_i) + O_p(h_0)$. So the expectation of the moment function of the SDML estimator $\gamma(U_i, X_i) + \frac{K_h(T_i - t)}{\hat{f}_\ell(t|X_i)} (Y_i - \gamma(U_i, X_i))$ over U_i given Z_i yields the moment function of the DML estimator in [\(1.1\)](#) as $h_0 \rightarrow 0$. [Assumption 4.1](#) gives the high-level conditions on the nuisance function estimators.

Assumption 4.1. For $\ell = 1, \dots, L$ and $t \in \mathcal{T}$, (a) $h_0^{-d_T/2} \|\hat{\gamma}_\ell - \bar{\gamma}\|_{F_{TX}} = o_p(1)$ and $\|\hat{f}_\ell - \bar{f}\|_{F_{tX}} = o_p(1)$; (b) $\sqrt{nh^{d_T} h_0^{-d_T/2}} \|\hat{\gamma}_\ell - \gamma\|_{F_{tX}} \|\hat{f}_\ell - f_{T|X}\|_{F_{tX}} = o_p(1)$; (c) $\|(\hat{\gamma}_\ell - \bar{\gamma})^2\|_{F_{TX}} = O_p(1)$; (d) Either $\bar{\gamma} = \gamma$ or $\bar{f} = f_{T|X}$.

Theorem 4.1 (SDML-Asymptotic normality). Let [Assumptions 2.1–2.3](#) and [4.1](#) hold. Let $h \rightarrow 0$, $nh^{d_T} \rightarrow \infty$, and $nh^{d_T+4} \rightarrow C \in [0, \infty)$. Let $h_0 \rightarrow 0$, $nh_0^{d_T} \rightarrow \infty$, and $\sqrt{nh^{d_T} h^2 h_0} \rightarrow 0$. Then the results in [Theorem 3.1](#) hold with $\check{\beta}_t$ in place of $\hat{\beta}_t$.

[Assumption 4.1\(a\)](#) strengthens the consistency condition in [Assumption 2.4\(a\)](#) with a penalty $h_0^{-d_T/2}$. Interestingly when $h_0 = h$, the rate condition (b) $\sqrt{n}\|\hat{\gamma}_\ell - \gamma\|_{F_{TX}} \|\hat{f}_\ell - f_{T|X}\|_{F_{tX}} = o_p(1)$, which is the same rate condition for the regular semi-parametric models in CCDDHNR. Note that when $h_0 \leq h$, [Assumption 4.1\(b\)](#) requires both nuisance functions are correctly specified or nonparametrically estimated, as discussed in [Remark 2.1](#). [Assumption 4.1\(c\)](#) is a boundedness condition that is implied if $\hat{\gamma}$ is uniformly bounded; for example, deep neural networks in FLM for a uniformly bounded Y .

Following the same approach for the DML estimator $\hat{\beta}_t$, we estimate the AMSE optimal bandwidth h_t^* given in [Theorem 3.2](#) by a data-driven bandwidth $\check{h}_t$ using the SDML estimator $\check{\beta}_t$. We present the consistency of $\check{h}_t$ in [Theorem S3](#) and the asymptotic theory for $\check{\theta}_t$ in [Theorem S2](#) in the appendix.

5. Numerical Examples

We provide numerical examples of Monte Carlo simulations and an empirical illustration. The estimation procedure of the proposed DML estimator is described in [Section 2](#). To estimate the first-step conditional expectation function $\gamma(t, x) = \mathbb{E}[Y|T = t, X = x]$ and the conditional density $f_{T|X}$ by MultiGPS as described in [Section 2.1](#), we employ three methods: Lasso, the deep neural networks (NN) based on Farrell, Liang, and Misra (2021b), and the Kernel NN proposed in [Section 3.1](#). We implement our DML estimator with these algorithms, respectively. Note that Lasso assumes certain sparsity specifications and NN is a nonparametric method. Our doubly robust inference theory allows one of the first-step functions to be misspecified. Software we develop is available at <https://github.com/KColangelo/Double-ML-Continuous-Treatment>.

5.1. Simulation Study

The data-generating process is $v \sim \mathcal{N}(0, 1)$, $\varepsilon \sim \mathcal{N}(0, 1)$, $X = (X_1, \dots, X_{100})' \sim \mathcal{N}(0, \Sigma)$, $T = \Phi(3X'\theta) + 0.75v - 0.5$, $Y = 1.2T + 1.2X'\theta + T^2 + TX_1 + \varepsilon\sqrt{0.5 + \Phi(X_1)}$, where $\theta_j = 1/j^2$, $\text{diag}(\Sigma) = 1$, the (i, j) -entry $\Sigma_{ij} = 0.5$ for $|i - j| = 1$ and $\Sigma_{ij} = 0$ for $|i - j| > 1$ for $i, j = 1, \dots, 100$, and Φ is the CDF of $\mathcal{N}(0, 1)$. This is a nonseparable model with the potential outcome $Y(t) = 1.2t + 1.2X'\theta + t^2 + tX_1 + \varepsilon\sqrt{0.5 + \Phi(X_1)}$. The parameters of interest are the average dose response function and the partial effect at $t = 0$, that is, $\beta_0 = \mathbb{E}[Y(0)] = 0$ and $\theta_0 = \partial \mathbb{E}[Y(t)] / \partial t|_{t=0} = 1.2$.

We compare estimations with fivefold cross-fitting ($L = 5$) and without cross-fitting ($L = 1$), and with a range of bandwidths to demonstrate robustness to bandwidth choice. We use the second-order Epanechnikov kernel with bandwidth h . For the MultiGPS estimator, we use the Gaussian kernel with bandwidth $h_1 = h$. Let the bandwidth $h = c\sigma_T n^{-0.2}$ for a constant $c \in \{1.0, 1.25, 1.5\}$ and the standard deviation σ_T of T . For $\hat{\theta}_0$, $\eta = hn^{-1/6}$. We computed the AMSE-optimal bandwidth h_0^* given in [Theorem 3.2](#) that has the corresponding $c^* = 1.44$. Thus, using some undersmoothing bandwidth with $c < c^*$, the 95% confidence interval $[\hat{\beta}_t \pm 1.96s.e.]$ is asymptotically valid, where the standard error (s.e.) is computed using the sample

Table 1. Simulation results for $\hat{\beta}_0$.

L	c	Lasso			Neural network			Kernel neural network		
		Bias	RMSE	Cov.	Bias	RMSE	Cov.	Bias	RMSE	Cov.
$n = 1000$										
1	1.00	0.008	0.100	0.935	0.064	0.108	0.890	0.037	0.111	0.689
	1.25	0.016	0.094	0.937	0.067	0.109	0.859	0.044	0.105	0.748
	1.50	0.025	0.091	0.925	0.077	0.112	0.840	0.051	0.103	0.778
5	1.00	0.008	0.100	0.936	−0.014	0.122	0.947	0.009	0.122	0.965
	1.25	0.016	0.094	0.936	−0.007	0.109	0.967	0.018	0.108	0.968
	1.50	0.025	0.092	0.925	0.002	0.100	0.969	0.029	0.102	0.956
$n = 10,000$										
1	1.00	0.005	0.036	0.956	0.005	0.032	0.957	0.006	0.037	0.885
	1.25	0.008	0.034	0.954	0.009	0.032	0.937	0.010	0.035	0.895
	1.50	0.012	0.033	0.944	0.014	0.032	0.918	0.014	0.034	0.899
5	1.00	0.005	0.036	0.953	0.014	0.039	0.948	0.006	0.039	0.950
	1.25	0.008	0.034	0.952	0.018	0.039	0.916	0.009	0.036	0.948
	1.50	0.012	0.033	0.944	0.023	0.039	0.899	0.014	0.035	0.945

NOTE: $L = 1$: no cross-fitting. $L = 5$: fivefold cross-fitting. The bandwidth $h = c\sigma_T n^{-0.2}$, and $c = 1.44$ for the AMSE-optimal bandwidth for β_t . The nominal coverage rate (Cov.) of the confidence interval is 0.95.

analogue of the estimated influence function, as described in Section 3.

Table 1 reports the results for $\hat{\beta}_0$ based on 1000 Monte Carlo replications. Table S2 in the appendix reports the results for $\hat{\theta}_0$. Under no cross-fitting ($L = 1$), the confidence intervals generally have lower coverage rates (Cov.) and the bias is larger than under cross-fitting. The estimators using NN and Kernel NN perform well in the case of fivefold cross-fitting, with coverage rates near the nominal 95%, especially in a smaller sample size $n = 1000$. Intuitively Kernel NN estimates the conditional expectation function $\gamma(t, x)$ locally at t , resulting in a smaller effective sample size nb^{d_T} than the full sample size n used by NN. Therefore, NN may outperform Kernel NN in small samples. The coverage rate and bias are improved the most for (Kernel) NN with cross-fitting, but only marginally for Lasso. As cross-fitting should improve our estimation in the case that the ML algorithm is over-fitting, it might suggest that Lasso does not have a severe over-fitting problem for this data-generating process. Overall these results demonstrate consistency with the theoretical results of this article, confirming the usefulness of cross-fitting for ML methods.

Section S2 in the appendix provides implementation details and additional results of SDML and ReGPS under two data-generating processes DGP1 and DGP2. We find that ReGPS has the following advantages over MultiGPS: (i) The ReGPS estimate is positive and bounded by construction, while MultiGPS has the logical flaw of possible negative estimates.¹² (ii) Under DGP1 with small GPS, ReGPS can outperform MultiGPS in terms of bias and coverage rates. In addition, under DGP2 with extremely small GPS where regular identification and asymptotic theory in Section 3 might not hold (Khan and Tamer 2010), ReGPS gives better bias and RMSE than MultiGPS.¹³ In sum, we suggest using ReGPS when MultiGPS estimates are negative or close

to zero. When the GPS is not small, ReGPS and MultiGPS are comparable. ReGPS might result in larger variance and more computational cost due to additional estimation.

5.2. Empirical Illustration

We illustrate our method by re-analyzing the Job Corps program in the United States, conducted in the mid-1990s. Job Corps is the largest publicly funded job training program, which targets disadvantaged youth. The participants are exposed to different numbers of actual hours of academic and vocational training. The participants' labor market outcomes may differ if they accumulate different amounts of human capital acquired through different lengths of exposure. We estimate the average dose response functions to investigate the relationship between employment and the length of training. As our analysis builds on Flores et al. (2012), Hsu et al. (2020), and Lee (2018), we refer the readers to the reference therein for further details of Job Corps. We use the same dataset in Hsu et al. (2020).

We consider the outcome variable (Y) to be the proportion of weeks employed in the second year following the program assignment. The continuous treatment variable (T) is the total hours spent in academic and vocational training in the first year. We follow the literature to assume the conditional independence Assumption 2.1(a), meaning that selection into different levels of the treatment is random, conditional on a rich set of observed covariates (X). The identifying Assumption 2.1 is indirectly assessed in Flores et al. (2012). Our sample consists of 4024 individuals who completed at least 40 hr (one week) of training. There are 40 covariates measured at the baseline survey. In the appendix, Figure S1 shows the distribution of T , and Table S8 provides brief descriptive statistics.

We estimate the average dose response function $\beta_t = \mathbb{E}[Y(t)]$ and partial effect $\theta_t = \partial \mathbb{E}[Y(t)] / \partial t$. We implement four DML estimators: Lasso, the generalized random forests in Athey, Tibshirani, and Wager (2019), the neural networks (NN) based on Farrell, Liang, and Misra (2021b), and the Kernel NN proposed

¹²We may adopt the trimming/flooring approaches to addressing this concern in the literature. For example, following Hsu, Lai, and Lieli (2020), we can use the estimate $\max\{\hat{\tau}_{TX}(t|X_i), \delta_n\}$ for some positive sequence $\delta_n \rightarrow 0$.

¹³Khan and Tamer (2010) study extensively *irregular identification* and inverse weight estimation, for example, the propensity score can be close to zero as a small denominator for the binary treatment case. If the GPS is extremely small as in DGP2, then the insight from the binary treatment literature suggests that the $\sqrt{nh^{d_T}}$ -convergence rate of our nonparametric $\hat{\beta}_t$ might

be also affected. Valid inference under such irregular identification is out of the scope of this article.

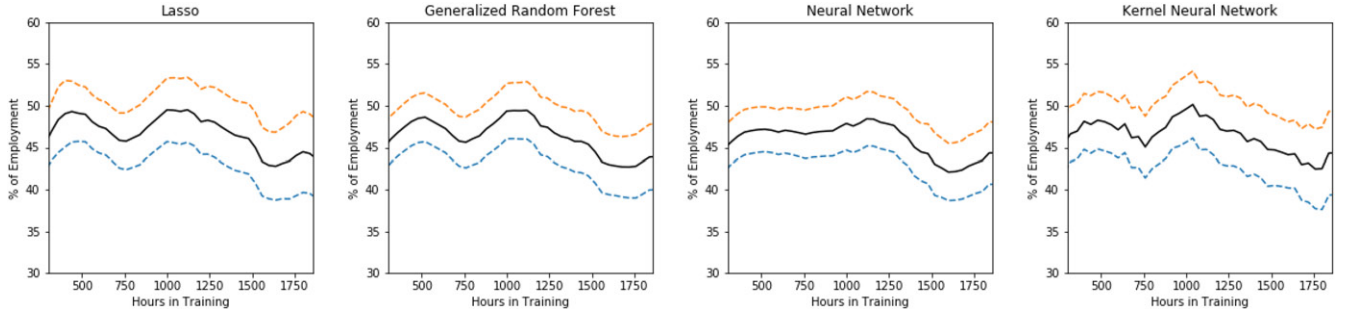


Figure 1. Estimated average dose response functions with 95% confidence intervals.

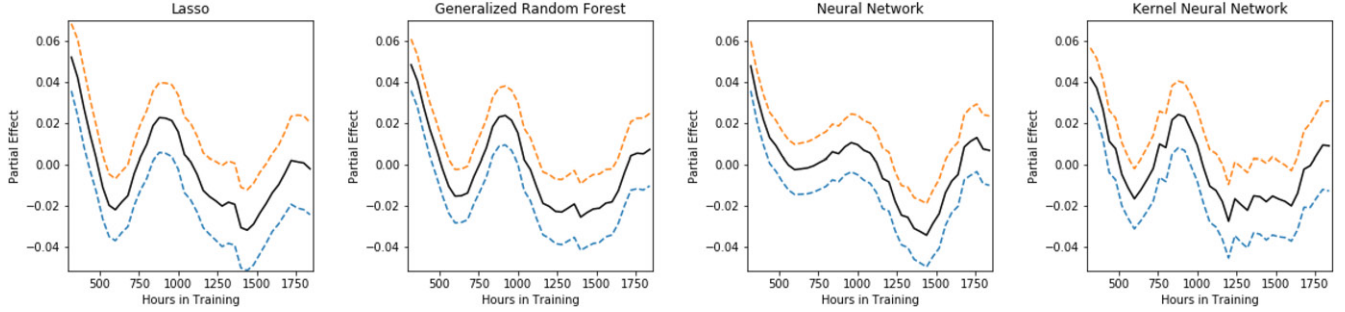


Figure 2. Estimated partial effects with 95% confidence intervals.

in Section 3.1. The parameters for these methods are selected as described in Section S2 in the appendix.

We compute the optimal bandwidth that minimizes an asymptotic integrated MSE. For practical implementation, consider a weight function $w(t) = \mathbf{1}\{t \in [\underline{t}, \bar{t}]\} / (\bar{t} - \underline{t})$ that is the density of Uniform $[\underline{t}, \bar{t}]$ on a subset of the support of T . We compute the bandwidth that minimizes the asymptotic integrated MSE $\int_{\mathcal{T}} (V_t / (nh^{d_T}) + h^4 B_t^2) w(t) dt$ for an integrable weight function $w(t) : \mathcal{T} \mapsto \mathcal{R}$ is $h_w^* = (d_T V_w / (4B_w))^{1/(d_T+4)} n^{-1/(d_T+4)}$, where $V_w \equiv \int_{\mathcal{T}} V_t w(t) dt$ and $B_w \equiv \int_{\mathcal{T}} B_t^2 w(t) dt$, following Theorem 3.2. We choose a finite set of m equally spaced grid points $[\underline{t}, \bar{t}] : \{t = t_1, t_2, \dots, t_m = \bar{t}\}$ for estimation. Following the approach given in Section 3, we estimate V_{t_j} with $h = 3\hat{\sigma}_T n^{-0.2} = 548.52$ and B_{t_j} with $b = 2h$ and $a = 0.5$, for $j = 1, \dots, m$. A plug-in estimator $\hat{h}_w^* = (\hat{V}_w / (4\hat{B}_w))^{1/5} n^{-1/5}$, where $\hat{V}_w = m^{-1} \sum_{j=1}^m \hat{V}_{t_j}$ and $\hat{B}_w = m^{-1} \sum_{j=1}^m \hat{B}_{t_j}^2$. We use $[\underline{t}, \bar{t}] = [160, 1840]$ and $t_j - t_{j-1} = 40$ in this empirical application. Then the under-smoothing bandwidths $0.8\hat{h}_w^*$ are 170.76 for Lasso, 179.47 for the generalized random forest, 179.4 for NN, and 180.09 for Kernel NN.

Figure 1 presents $\hat{\beta}_t$ along with 95% point-wise confidence intervals. The results have similar patterns across the ML nuisance function estimators and suggest an inverted-U relationship between the employment and the length of participation.

Figure 2 reports the partial effect estimates $\hat{\theta}_t$ with step size $\eta = 160$ (one month). Across all procedures, we see positive partial effects when hours of training are less than around 500 (three months) and negative partial effect around 1500 hr (9 months). Taking the estimates by NN for example, $\hat{\beta}_{400} = 46.83$ with standard error $s.e. = 1.38$ and $\hat{\theta}_{400} = 0.0217$ with $s.e. = 0.0123$. This estimate implies that increasing the training from two months to three months increases the average proportion of

weeks employed in the second year by 3.47% (nearly two weeks) with $s.e. = 1.962\%$.

6. Conclusion and Outlook

Our DML estimator serves as the preliminary element for policy learning and optimization with a continuous decision, following Manski (2004), Hirano and Porter (2009), Kitagawa and Tetenov (2018), Kallus and Zhou (2018), Demirel et al. (2019), Athey and Wager (2019), among others. Another extension is robustness against multiway clustering, where the conventional cross-fitting does not ensure the independence between observations in I_ℓ from I_ℓ^c , following Chiang et al. (2021) for regular DML estimators.

Supplementary Materials

The supplementary file contains the asymptotic theory for the partial effects, proofs, and additional Monte Carlo simulations, kernel localization, the kernel and series nuisance function estimators, and uniform inference theory.

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Disclosure Statement

The author reports there are no competing interests to declare.

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